Journal of Economic Criminology 7 (2025) 100127

Contents lists available at ScienceDirect

Journal of Economic Criminology

journal homepage: www.journals.elsevier.com/journal-of-economic-criminology

Harnessing artificial intelligence and machine learning for fraud detection and prevention in Nigeria

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ARTICLE INFO

Keywords:
Artificial Intelligence
Machine learning
Fraud

ABSTRACT

Fraud poses a significant threat to Nigeria's burgeoning digital economy, impacting sectors like finance, e-commerce, healthcare, and education. Traditional methods struggle to keep pace with evolving fraud schemes. This paper investigates the potential of Artificial Intelligence (AI) and Machine Learning (ML) for enhanced fraud detection and prevention in Nigeria. We explore various AI methodologies, including supervised, unsupervised, and deep learning. We discuss their applications in anomaly detection, behavioural analysis, risk scoring, and network analysis. By leveraging AI's continuous learning capabilities, organizations can adapt to novel fraud tactics. The paper highlights the benefits of AI-powered fraud detection, including increased efficiency, improved accuracy, and proactive risk mitigation. However, challenges like technical limitations and regulatory considerations are acknowledged. Lastly, we examine the promising future of AI and ML in transforming the financial crime prevention situation in Nigeria.

1. Introduction

The digital revolution has undeniably transformed Nigeria, encouraging increased connectedness and accelerating growth. From e-commerce and digital banking to e-learning and telemedicine, the digital revolution is empowering Nigerians with tools to bridge infrastructural gaps, access global opportunities, and drive sustainable growth across various sectors (Okonkwo et al., 2024).

However, despite this transformative impact of the digital revolution in Nigeria, it has also given rise to a significant challenge: an increase in fraud in several industries. Fraud has become a serious concern, endangering not only businesses but also the country's financial stability, affecting everything from e-commerce and healthcare to education. The consequences of these crimes go far beyond the lives of the specific victims; they affect the basis of the Nigerian economy itself. Companies suffer catastrophic losses, customer confidence declines, and there is a general decline in the perception of security associated with digital transactions (Okoye and Gbegi, 2013). This worrying pattern presents a worrisome picture. Global reports on fintech and payments show that between 2023 and 2027, online payment fraud alone is projected to cost firms globally more than $343 billion (Juniper Research, 2022). In Nigeria, the effects are felt not only by large organizations but also by regular people who become victims of fake online markets, identity theft, and various other forms of deception (Oluwole, 2024). The complexity of fraudsters is simply surpassing that of traditional systems of fraud detection, which rely on strict standards and manual checks. A paradigm change is required at this crucial juncture to create a new line of defense that is powered by artificial intelligence (AI) and machine learning (ML) (Kamuangu, 2024).

Artificial Intelligence is the capability of machines to imitate intelligence in human behaviour, attained by examining how the human brain thinks and how it learns, decides, and works in solving a specific problem (Abhulimen and Ogunti, 2021). AI aims to replicate these capabilities in machines, enabling them to perform tasks that typically require human intelligence, such as reasoning, understanding language, recognizing patterns, and adapting to new inputs. AI can take various forms, including expert systems, speech recognition tools, and advanced robotics, and it is driven by methods that enable machines to act intelligently, even in complex or unpredictable situations.

Machine learning, on the other hand, is a subset of AI that uses algorithms that analyze large amounts of data, identify patterns, and make predictions or decisions based on that information (Sarker, 2021). For instance, ML algorithms are designed to improve their accuracy and efficiency over time as they are exposed to more data. Unlike traditional programming, where developers write explicit rules, ML systems "learn" these rules automatically through training processes.

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https://doi.org/10.1016/j.jeconc.2025.100127
Received 21 October 2024; Received in revised form 1 January 2025; Accepted 6 January 2025

AI and machine learning developments are revolutionizing the fight against fraud and provide a glimmer of hope in the fight against these crimes. Real-time pattern recognition and data analysis are possible for AI systems because of their capacity to emulate human cognitive functions. Because of this, they can detect suspicious activities that would go undetected by conventional means. These computers may identify irregularities that may indicate possible fraud attempts in a matter of seconds by filtering through enormous datasets at an astonishingly fast pace (Gupta, 2023).

This paper looks into how AI and ML might be revolutionary tools in the fight against fraud, particularly in Nigeria. We will examine how companies and organizations across a variety of sectors, such as banking, e-commerce, healthcare, and education, may use these technologies to detect and prevent fraud. By evaluating massive volumes of data, spotting intricate trends, and making real-time modifications that will result in a more lucrative and safer digital environment for Nigerians, AI and ML offer a cutting-edge defense against the schemes of fraudsters.

Both primary and secondary data were used to explore the role of AI and ML in fraud detection and prevention in Nigeria. Secondary data was sourced from various credible academic publications, industry reports, government records, and global fraud studies, including datasets on fraud trends, technological adoption, and sectorial impacts. Primary data, though limited, consisted of expert insights obtained through unstructured interviews with stakeholders in the Nigerian banking and finance, e-commerce, healthcare, and educational sectors. Additionally, anecdotal evidence based on our experiences as residents in Nigeria provided contextual understanding. These sources provided a foundation for understanding the broader context of fraud in Nigeria, the effectiveness of AI and ML technologies globally, and the practical challenges and adoption barriers.

2. Economic impact of fraud in Nigeria across various sectors

The insidious nature of fraud extends far beyond the immediate financial losses inflicted upon individuals and businesses. In a nation like Nigeria, striving for economic prosperity and digital inclusion, fraud acts as a pernicious force, eroding trust, hindering growth, and stifling innovation (Drammeh, 2023). This chapter explores how fraud affects the economy in many different ways, including banking and finance, e-commerce, healthcare, and education. It also emphasizes how urgently artificial intelligence (AI) and machine learning (ML) are needed as revolutionary countermeasures.

2.1. Banking and finance

Manyo et al. (2023), in their research, revealed that the financial sector is a frequent target for fraudsters, with consequences extending beyond immediate financial losses. The research further reveals that banks facing fraudulent transactions suffer significant financial setbacks, which negatively affect their profitability and ability to invest in crucial services. This disruption affects the flow of credit to both businesses and individuals, slowing down economic growth. Additionally, the rise in financial fraud undermines public trust in the banking system, discouraging the use of digital financial services and thereby impeding the government's efforts to promote financial inclusion, which is critical for poverty reduction and economic development.

A report by Akintaro (2023) highlighted the scale of the problem in Nigeria, where 24 commercial banks lost N5.79 billion to fraud in the second quarter (Q2) of 2023 - an increase of 1125 % from the N472 million lost in the first quarter (Q1) of the same year. The total value of fraudulent transactions also surged dramatically from N2.58 billion to N9.75 billion, increasing by nearly 278 % from Q1 to Q2. Fraudulent loans accounted for 94.35 % of these losses, revealing a serious flaw in the Nigerian banking system. These fraudulent loans divert funds from honest businesses and entrepreneurs, stifling new business formation and job creation, which are crucial to economic development.

The report (Akintaro, 2023) also reveals that other types of fraud, including mobile and computer fraud, pose additional threats. Mobile fraud, responsible for 3.39 % of the total losses, undermines trust in mobile banking platforms, a key driver of financial inclusion. Computer and web fraud, amounting to N59.5 million, highlights the ongoing vulnerability of online banking systems, deterring both consumers and businesses from engaging with digital services, thus hindering the growth of the digital economy.

2.2. E-commerce

E-commerce in Nigeria has the potential to drive economic growth, foster innovation, and open new opportunities. However, fraud on payment platforms presents a significant barrier to this growth. Recent increases in fraud have made the e-commerce environment fearful, causing consumers to be hesitant about online purchases due to concerns about payment security. This reluctance leads to a preference for cash-on-delivery, slowing the government's push for a cashless economy (Okolie and Ojomo, 2020).

E-commerce fraud comes in many forms, affecting both consumers and merchants. Consumers face risks like account takeover, mobile SIM switch fraud, and social engineering tactics such as phishing. These threats damage their trust in online transactions and increase the risk of financial loss (Komandla, 2024). Merchants, on the other hand, encounter fraud like identity theft, chargeback fraud, and man-in-the-middle attacks, which result in financial losses and discourage other businesses from entering the e-commerce space (Mutemi and Bacao, 2024).

Other types of fraud, such as card testing, refund fraud, friendly fraud, triangulation fraud and interception fraud add to the complexity of the problem. Card testing uses stolen credit card information for small purchases to test its validity, while refund fraud exploits stolen payment information to claim refunds for fake overpayments. Friendly fraud occurs when customers falsely claim they didn’t receive their goods to get a refund. Triangulation fraud involves fake online stores stealing customer information to make purchases from legitimate merchants. Interception fraud occurs when a person uses a stolen card with the provided address and billing information to make a purchase, then intercepts the package for themselves (AxiomQ, 2022).

The economic impact of e-commerce fraud in Nigeria is clear. It deters online purchases, hampers the growth of cashless transactions, and creates a climate of distrust. Merchants face significant financial risks and become hesitant to continue in the industry, stalling the sector's growth potential.

Projections indicates a significant increase in financial losses over the coming years. It is expected to escalate globally from $44.3 billion in 2024 to $107 billion by 2029, marking a 14 % increase (Adewumi, 2024). The rapid growth of Nigeria’s e-commerce market presents an increased opportunity for E-commerce fraudsters (Oloni, 2024)

2.3. Healthcare

Healthcare fraud in Nigeria is a significant issue with serious economic and social impacts, undermining the quality of care and the integrity of the healthcare system. Medical identity theft and fraudulent claims divert resources from patients who genuinely need care. This leads to compromised healthcare delivery and creates inefficiencies in the system (Akokuwebe and Idemudia, 2023).

A common form of fraud involves submitting inaccurate or misleading information to health insurance companies, tricking them into paying for unapproved benefits. In Nigeria, healthcare providers often engage in double billing, charging for services multiple times, or billing for treatments that were never received (Yange, 2019). This practice burdens the National Health Insurance Scheme (NHIS), increasing costs and reducing the effectiveness of care.

'Ghost patients' are another fraudulent tactic where claims are filed for individuals who do not exist. In addition, stolen identities are used to submit fraudulent claims, adding further complexity to the issue (Thornton et al., 2015). There is also a risk of collusion between patients and healthcare professionals, where bribes are exchanged to facilitate false claims.

These fraudulent activities result in increased administrative costs and reduced profitability for insurance companies, ultimately leading to higher premiums for average Nigerians. Patients suffer too, as false claims deplete the funds needed for essential healthcare infrastructure, resulting in reduced access to quality care for those in need.

2.4. Education

The Nigerian education sector faces significant challenges from examination fraud falsified scholarship applications, and fake credentials, all of which erode public trust and weaken the integrity of the system. These dishonest practices harm the quality of education and workforce readiness, denying opportunities to deserving students while benefiting unqualified individuals. As a result, economic growth and innovation are stifled due to an inadequately trained workforce (Alordiah et al., 2023).

Examination fraud undermines fair evaluation, from simple cheating to bribing invigilators. Students who cheat gain an unfair advantage, damaging the reputation of educational institutions and leading to a generation ill-prepared for their careers. Employers, frustrated by unreliable academic credentials, struggle to hire competent workers, further hindering economic progress (Eneji et al., 2022).

Admission fraud is another concern, where students gain university entry through bribes or forged documents, bypassing academic merit. This weakens the intellectual standard of institutions, diminishes opportunities for deserving students, and results in unqualified graduates who are ill-equipped for the workforce (Noah and Adesope, 2022) This not only affects their financial prospects but also poses risks to public safety in fields that require technical expertise.

Contract cheating, where students outsource their academic work to third-party services (Ali, and Alhassan, 2021), threatens the learning process by allowing students to "buy" their achievements. This erodes the development of critical thinking and knowledge acquisition, leading to a workforce deficient in these essential skills, further stalling economic progress.

Lastly, teacher fraud, though less common, also contributes to academic decline. Teachers may inflate grades or use fake credentials to secure employment, compromising the quality of teachers (Akinniyi et al., 2021). This deprives students of necessary skills, affecting their future success and reducing the overall productivity of the workforce.

Addressing these forms of educational fraud through strict measures and fostering a culture of academic honesty is crucial to improving Nigeria's education system, workforce, and economy.

3. Traditional method of fraud detection

To safeguard their operations and customers, organizations have traditionally relied on a combination of pre-defined rules and manual analysis for fraud detection. While these methods have served a purpose in the past, they are increasingly proving inadequate in the face of ever-evolving threats.

Limitations in terms of adaptability are among the biggest problems with traditional fraud detection techniques (Bello et al., 2024). Usually, these systems work with a predetermined set of rules that identify actions that meet predetermined standards or surpass predetermined bounds. These regulations may have worked well in the past, but they are unable to keep up with how fraud is changing. To get beyond these pre-established red flags, fraudsters are always coming up with fresh ideas and strategies. It is frequently a time-consuming and slow procedure to update these rule-based systems to handle new threats (Liu et al., 2021). There is a gap in the detection rules that allows for the introduction of new fraud strategies, which provides an opportunity for fraudsters to take advantage of (Takyar, 2024). A fraudster might easily get around an e-commerce platform regulation that flags transactions over a particular amount as possibly fraudulent by breaking up a large purchase into several smaller transactions, each of which falls below the threshold and so avoids detection.

Traditional methods also suffer from a lack of sophistication in pattern recognition. Fraudulent operations may involve complex links and connections between different data points (Hilal et al., 2022). The use of a new user account, a billing address that is different from the shipping address, and an attempt to buy a remote location are a few examples of the various components of a fraudulent transaction. Hand analysis or rule-based algorithms that focus on individual data points alone could overlook these subtle but significant patterns.

Finally, traditional methods face significant scalability challenges. An explosion of data characterizes the digital age (Theodorakopoulos et al., 2024). The amount of data points, user interactions, and online transactions is growing constantly, making human analysis of this data more challenging and time-consuming. Traditional methods may be ill-suited to effectively handle these enormous datasets. This may result in an accumulation of unprocessed data, which could give rise to chances for fraudulent operations to evade detection.

While traditional fraud detection methods have played a historical role in safeguarding organizations, their limitations are becoming increasingly apparent in today's dynamic digital landscape. Their inability to adapt to evolving threats, identify complex patterns, and handle large datasets creates vulnerabilities that fraudsters can exploit (Bansal et al., 2024). As we move forward, there is a critical need for more robust and adaptable solutions to combat fraud.

4. The urgency for AI intervention

Fraudulent actions undermine confidence, divert resources, and inhibit innovation in a variety of sectors, including financial organizations, e-commerce platforms, healthcare providers, and educational institutions. At this pivotal moment, machine learning (ML) and artificial intelligence (AI) emerge as game-changing solutions to tackle these ubiquitous problems (Sharma et al., 2024).

Traditional methods of fraud detection have proven increasingly inadequate. Manual processes are slow, error-prone, and overwhelmed by the ever-evolving tactics of fraudsters. Legacy systems struggle to identify complex patterns hidden within vast amounts of data. (Hilal et al., 2022). This is where AI and ML offer a powerful solution.

AI's ability to analyze massive datasets and identify subtle anomalies makes it ideal for fraud detection. Machine learning algorithms can be trained on historical fraud cases to recognize patterns and predict future fraudulent activities. This allows for real-time monitoring of transactions, flagging suspicious activity before it can cause financial harm (Bansal et al., 2024).

In the banking and finance sector, AI can analyze customer behavior, spending patterns, and account activity to identify potential fraudulent loans, unauthorized access attempts, and money laundering schemes (Hassan et al., 2023). This empowers financial institutions to take proactive measures, such as account freezes or additional authentication steps, to mitigate losses.

The e-commerce sector stands to gain immensely from AI-powered fraud detection. AI can analyze customer purchase history, IP addresses, and device fingerprints to identify suspicious orders that may be linked to stolen credit card information (Gayam, 2020). Additionally, AI can be used to detect fake reviews and identify fraudulent merchants operating on e-commerce platforms (Paul and Nikolaev, 2021).

The healthcare sector can leverage AI to combat fraudulent claims and identity theft. AI algorithms can analyze medical records, billing data, and patient information to identify anomalies that may indicate fraudulent activity (Kasaraneni, 2024) This can help healthcare providers and insurance companies prevent misuse of resources and ensure that funds are directed towards genuine medical needs.

The education sector can utilize AI to combat examination malpractice and admission fraud. AI-powered proctoring systems can detect suspicious behavior during online exams (Nagpal et al., 2024), while advanced document verification tools can identify forged certificates and transcripts (Boonkrong, 2024). Furthermore, AI can analyze student performance data to flag potential plagiarism in assignments and essays (Tripathi and Thakar, 2024).

The urgency for AI intervention in combating fraud cannot be overstated. By leveraging the power of AI and ML, Nigeria can safeguard its economy, build trust within critical sectors, and empower its citizens to participate with confidence in a digital future. As AI continues to evolve, its capabilities in fraud detection will only become more sophisticated, paving the way for a more secure and prosperous Nigeria.

5. AI and machine learning approaches for effective fraud detection and prevention

While AI provides the broad framework for intelligent systems, Machine Learning (ML) serves as the engine that drives its effectiveness in fraud detection. Machine Learning (ML) is a subset of AI that empowers machines to identify patterns and improve their performance on a specific task without explicit programming (Sarker, 2021). There are three main categories of Machine Learning algorithms particularly well-suited to combatting fraud in Nigeria's digital sectors.

5.1. Supervised learning

One way to conceptualize supervised learning is as a pupil carefully studying under an instructor. The teacher is the labelled data in this comparison, where each data point has a label that corresponds to its category (fraudulent or valid transaction, for example). For example, financial institutions can use transaction data from the past to train models that identify trends linked to fraud. Like the student, these models are always picking up new skills and honing the ones they already have in recognizing patterns. Supervised learning-enabled AI systems can evaluate fresh transactions in real-time and mark those with questionable patterns for additional examination (Afriyie et al., 2023). This makes it possible to take preventative action, which may stop financial losses before they happen.

5.2. Unsupervised learning

Unsupervised learning takes a different approach. It is particularly valuable for detecting novel fraud schemes that do not conform to known patterns. Anomaly detection, a common unsupervised learning technique, identifies outliers or deviations from the norm in a dataset (Hilal et al., 2022). In the context of fraud detection, this can be used to uncover unusual or unexpected behavior, such as unusually large transactions originating from unfamiliar locations or sudden spikes in account activity (Rai and Dwivedi, 2020). Imagine a scenario where a customer with a history of modest online purchases suddenly attempts to purchase a high-end electronic device from a new location. An unsupervised learning model can detect this deviation from the norm and flag it for investigation, potentially preventing a fraudulent purchase.

5.3. Deep Learning

Deep learning, a powerful subset of ML, has gained prominence for its ability to process vast amounts of data and extract intricate patterns. Neural networks, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), are at the core of deep learning (Chen and Lai, 2021). These complex algorithms function similarly to the interconnected neurons in the human brain, allowing them to learn and adapt with remarkable efficiency. CNNs excel at image recognition, making them ideal for analyzing images of signatures, checks, or identification documents to identify forgeries or alterations used in fraudulent attempts (Emiroğlu, 2023). Imagine a CNN analyzing a scanned copy of a driver's license. By recognizing subtle inconsistencies in the document's formatting or identifying signs of tampering, CNN can flag the document as potentially fraudulent, helping to prevent identity theft. RNNs, on the other hand, are adept at sequential data analysis, making them valuable for fraud detection in financial transactions where the order and timing of events can be crucial for identifying suspicious activity (Bello et al., 2022). An RNN can analyze a series of bank transactions, paying close attention to the sequence and timing of each event. By identifying unusual patterns, such as a rapid succession of withdrawals from geographically distant ATMs, the RNN can flag the account for potentially fraudulent activity.

6. Some common applications of machine learning in fraud detection and prevention

The following are some common applications of machine learning in fraud detection and prevention:

6.1. User authentication and biometric recognition

Traditional password-based authentication systems are susceptible to brute-force attacks and social engineering. Machine learning (ML) algorithms can significantly enhance security by analyzing user behavior and biometric data (Castelblanco et al., 2020). These algorithms can learn a user's typical login patterns, including factors like the device used, location, and time of day. Deviations from this established baseline can trigger alerts for potential unauthorized access attempts. Biometric recognition, such as fingerprint or facial recognition, adds another layer of security. ML can be used to analyze and verify biometric data in real-time, ensuring only authorized users gain access. Arslan et al. (2016) explored the application of ML in biometric authentication systems, emphasizing its role in user verification and security. Their review highlights successful implementations of machine learning algorithms like Support Vector Machines (SVMs), Naive Bayes (NB), and Artificial Neural Networks (ANNs) in achieving high recognition rates (80–99 %) and also emphasizing the potential of ML for liveness detection and spoofing prevention which is a crucial aspect for robust user authentication and fraud mitigation in financial institutions. The limitations of traditional passwords were recognized by Nnamoko et al. (2022) thereby presenting a compelling case for behavioral biometrics as a machine learning application to enhance fraud detection and prevention in online transactions. They focused on leveraging users' inherent behavioral patterns, such as keystroke dynamics and mouse movements, for user identification, aiming to improve authentication security and combat identity theft in the financial sector. Abhulimen and Ogunti (2022) explored the use of residual convolutional neural networks for age estimation in a possible attempt to curb age falsification crime with face images. Yousefi et al. (2019) also explored the convergence of machine learning and user authentication methods specifically for credit card fraud detection in a survey highlighting the limitations of static password-based authentication and exploring how behavioral biometrics, like touch dynamics and signature patterns, can be leveraged with machine learning to create more robust user authentication schemes. Machine learning's ability to analyze biometric data offers an additional layer of security beyond traditional passwords or PINs, effectively distinguishing legitimate users from imposters attempting fraudulent transactions. This aligns perfectly with Nigeria's growing need for secure biometric identification systems to combat fraud across various sectors.

6.2. Behavior analysis

Customer behavior analysis is a powerful tool for fraud detection. ML algorithms can analyze a user's historical transaction data, spending habits, and online activity to establish a baseline for their typical behavior (Jóhannsson, 2017). Deviations from this baseline, such as sudden spikes in purchase amounts, unusual purchases from unfamiliar locations, or a rapid increase in login attempts from geographically distant locations, can all be red flags for potential fraudulent activity. Sadgali et al. (2021) offered a behavior analysis approach that uses cardholder behavior to identify credit card fraud. Their methodology makes use of a hybrid machine learning strategy that combines fuzzy logic, association rule learning, and rough set theory to analyze user behavior and produce rules for spotting fraudulent transactions while maintaining low processing times. For health insurance fraud detection, Lu et al. (2023a, 2023b) introduced a noteworthy behavioral analysis model, MHAMFD using attributed heterogeneous information networks (AHINs), to capture the complex interactions that exist between physicians, hospitals, patients, and other healthcare providers. This methodology incorporates behavioral linkages across visits, going beyond individual patient data analysis and by giving priority to the most pertinent relationships, MHAMFD's hierarchical attention mechanism improves the accuracy of fraud detection and provides comprehensible results that can direct the inquiries of healthcare practitioners. By continuously monitoring behavior patterns, ML systems can identify anomalies and flag suspicious accounts for further investigation. This proactive approach allows organizations to take preventive measures before fraudulent transactions occur.

6.3. Anomaly detection

Machine learning excels at anomaly detection, making it a powerful tool for fraud prevention. Anomaly detection algorithms can analyze various amounts of data in real-time, searching for deviations from established patterns that might indicate fraudulent activity. These patterns could include unusual transaction amounts, a sudden increase in activity from a dormant account, or multiple login attempts from geographically disparate locations in a short timeframe (Hilal et al., 2022). Researchers are exploring different machine learning techniques to enhance fraud detection in this area. For instance, Patrik et al. (2020) proposed a system that combines machine learning's strength in identifying suspicious transactions with the power of graph databases to unearth connections between data points in an approach addressing the limitations of standalone machine learning methods, which might miss crucial relationships within the data, to achieved a high accuracy rate in fraud detection. Another is the research by Okechukwu et al. (2023) which utilizes a deep learning technique called a feed-forward neural network to achieve a 97 % detection rate for fraudulent credit card transactions and phishing websites, demonstrating machine learning's effectiveness in identifying anomalies that deviate from normal banking activities. Anomaly detection in financial statements of Vietnamese listed companies was investigated in a paper by Lokanan et al. (2019) that not only confirm the effectiveness of machine learning in uncovering irregularities but also suggest its potential for ranking companies based on their financial health, particularly valuable in developing economies. Real-time monitoring allows organizations to swiftly identify and respond to potential fraud attempts, minimizing losses. When anomalies are detected, the system can alert relevant personnel for further investigation.

6.4. Risk assessment

ML can be harnessed to create robust risk assessment models to analyze various factors associated with a transaction, such as the amount, location, type of purchase, and the user's past behavior (Fraud.net, 2024). By assigning a risk score to each transaction, the system can prioritize which transactions require further scrutiny. One approach, explored by Guan et al. (2023), combines ML with human expertise to strengthen credit risk assessments in a that research demonstrates that ML models, particularly when trained on limited data, can benefit from the addition of Ripple-Down Rules (RDR) created by a domain expert. These RDRs capture valuable insights that can improve the model's accuracy and fairness, especially in handling cases like fraud detection and prevention. A thorough framework using five different algorithms (Logistic Regression, SVM, XGBoost, GAN with autoencoder, and OCSVM) was developed by Lu et al. (2023a, 2023b) in research investigating machine learning's potential for corporate fraud detection in China. The merits of machine learning in risk assessment are reinforced by their findings which shows supervised learning techniques to be beneficial, as demonstrated by XGBoost's remarkable accuracy in spotting fraudulent organizations. Using such techniques for risk assessment in fraud detection efforts is made more compelling by the alignment of machine learning algorithms with pertinent financial data. Risk assessment in credit transactions was examined by Malik et al. (2024), in a research that detailed the benefits of machine learning (ML) algorithms fraud detection and using large datasets to produce faster, more accurate judgments while analyzing the drawbacks of conventional approaches. Several machine learning models, such as Support Vector Machines, Random Forests, and Logistic Regression, show how good they are at assessing credit risk, were studied. They also dug into feature engineering strategies for extracting useful information from financial data, which adds significantly to ML and may be used to improve risk assessment procedures and create a more secure financial environment. This risk-based approach will allow organizations in Nigeria to allocate resources efficiently. High-risk transactions can be flagged for immediate investigation, while low-risk transactions can be processed smoothly, minimizing disruption for legitimate users.

6.5. Text analysis

The digital world generates a vast amount of text data, including emails, social media posts, and customer reviews. ML algorithms can be employed to analyze this unstructured data and identify patterns or keywords that might indicate fraud or scams (Goyal et al., 2020). For example, ML can detect suspicious language or phrasing often used in phishing attempts, protecting unsuspecting users from falling victim to these deceptive tactics. Phua et al. (2010) explored the use of text analysis and machine learning to detect fraudulent emails using a dataset of emails classified as valid or spam, employing algorithms such as Support Vector Machines (SVM), Random Forest, Naive Bayes, and Decision Tree classifiers. According to the results, SVM had the highest accuracy, hitting over 98 % and this shows how text analysis, enabled by machine learning method, may greatly enhance the detection of fraud in email correspondence, which is a common medium for fraudulent activities. Oladimeji (2019) also investigated the application of machine learning for phishing email detection in Nigeria and suggested methods such as Naive Bayes and Support Vector Machines with the use of pre-processing techniques such as word embedding and stemming to categorize phishing attempts with high accuracy. This study opens the door for more investigation into the use of machine learning in Nigeria to counteract different types of fraud attempts by demonstrating the technology's potential for textual analysis-based fraud detection. Text analysis applications can also be seen in financial data in research by Li et al. (2023), in China examining the effectiveness of ML in analyzing a firm's Management Discussion and Analysis (MDA) section for fraud detection and highlighting the potential of text analysis. This paper demonstrated how machine learning (ML) may be used to analyze sentiment, readability, and forward-looking statements in addition to financial data and also to show that when textual analysis is added to financial measures, fraud detection algorithms perform better. This Chinese study suggests that textual data analysis from financial reports and communications will serve as a potent tool for fraud identification and prevention. These insightful findings apply to the Nigerian situation.

7. Benefits of AI and ML in fraud detection

Fraudsters are continuously coming up with new plans, finding ways around restrictions, and taking advantage of regulation gaps. The problem is made even more difficult by the enormous amount of data produced in today's digital environment. When handling large datasets, manual analysis becomes unreliable and ineffective, making organizations susceptible to fraudulent activity that goes unnoticed (Reurink, 2018). By leveraging AI and ML, organizations may create resilient and flexible fraud detection systems to protect their operations.

One of the most compelling benefits of AI and ML in fraud detection lies in their ability to perform real-time analysis. Unlike traditional methods that rely on historical data analysis, AI systems can continuously analyze data streams as they occur. This allows for the immediate identification of anomalies in real time (Bansal et al., 2024). Imagine a scenario where a fraudster attempts to make an unauthorized withdrawal from a bank account. An AI-powered system, constantly monitoring transaction activity, can detect this anomaly in real-time and trigger an alert. This swift intervention can potentially prevent the fraudulent transaction from being completed, safeguarding the account holder's funds.

Furthermore, AI excels in handling scalability. An explosion of data characterizes the digital age (Theodorakopoulos et al., 2024). Financial institutions, e-commerce platforms, and other sectors generate massive datasets daily. Manually analyzing such vast quantities of data for potential fraud is not only time-consuming but also prone to human error. AI systems, on the other hand, can effortlessly process and analyze these large datasets, identifying hidden patterns and anomalies that might escape human scrutiny (Khurana, 2020). This comprehensive analysis strengthens fraud detection efforts, leading to a more secure digital environment for all stakeholders.

Another significant advantage of AI and ML is their ability to reduce false positives (Bansal et al., 2024). Conventional fraud detection techniques frequently result in a large percentage of false positives or transactions that are lawful but are incorrectly marked as suspicious. This interferes with user experiences in addition to wasting expensive resources. However, to reduce false positives, AI algorithms can be continuously trained and improved. Artificial intelligence (AI) systems grow proficient at differentiating between authentic and fraudulent behavior by absorbing fresh information on fraudulent behaviors and learning from previous encounters. The utilization of a targeted strategy lessens the workload for human analysts, allowing them to concentrate their skills on examining the most crucial instances. This results in a fraud detection system that is more effective and efficient.

The potential of AI and ML to reveal hidden patterns is perhaps its most exciting feature (Amiri et al., 2024). Even with their great skill, human analysts are not immune to the subtle, complicated patterns hidden in large datasets. AI, on the other hand, is remarkably good at recognizing patterns. Artificial intelligence (AI) systems can spot minute links and connections in data that humans might miss. This makes it possible to identify fraudulent activity types that were previously unidentified, allowing for the proactive adoption of countermeasures before the development of widespread risks. Consider the scenario where an artificial intelligence system discovers a new phishing scheme aimed at gullible internet buyers. AI can protect the integrity of the e-commerce ecosystem and stop many people from becoming victims by recognizing the telltale signals of this fraudulent scam (Bansal et al., 2024).

Real-time analysis, scalability, reduced false positives, and the ability to uncover hidden patterns are just a few of the compelling benefits that AI and ML bring to the table. By embracing these advancements, Nigeria can create a more secure and trustworthy digital environment, encouraging economic growth and empowering its citizens to participate confidently in the digital age.

8. Challenges in AI-powered fraud detection systems

Although machine learning (ML) and artificial intelligence (AI) provide a potent tool in the battle against fraud, there are obstacles in the way of their application. When implementing AI-powered fraud detection systems, organizations need to be ready to face a variety of challenges from technological constraints to legal issues.

The availability and quality of data present one difficulty (Lin, 2024). High-quality and pertinent data is essential for AI systems to train and improve their detection models. Inaccurate, out-of-date, or incomplete data can seriously impair these algorithms' performance. The matter is made more complex by privacy concerns and data rules, which may restrict the number of extensive datasets that are available for AI systems to use for learning (Mothukuri et al., 2021). Organizations need to become adept at striking a careful balance between protecting access to essential information, guaranteeing data integrity, and adhering to privacy rules.

Another challenge is integrating AI fraud detection solutions with the current infrastructure (Bello and Olufemi, 2024). Modern AI and ML technologies might not be easily integrated with legacy systems, requiring major modifications or even total rebuilds. During the transition phase, there may be downtime or decreased functionality due to this resource-intensive and disruptive integration procedure. AI system integration requires meticulous planning and execution on the part of organizations to minimize interruptions and guarantee a seamless transition (Challoumis, 2024).

Another big obstacle is that fraud is a constantly changing field. Fraudsters are always coming up with new strategies and plans to get around detection systems (Hilal et al., 2022). No matter how sophisticated, AI models need to be updated frequently with the most recent information on fraudulent activity to be successful. This means that continual training of the AI system is required to keep it flexible and proactive in the battle against fraud. AI systems that are static and non-adaptive become vulnerable when scammers create new methods (Adhikari et al., 2024).

Beyond moral and legal obligations, there are other barriers to efficient AI-powered fraud detection. Ensuring compliance with relevant regulations, such as privacy and data protection statutes like the Nigeria Data Protection Regulation (NDPR), is imperative for organizations (BanwoandIghodalo, 2023). When AI is used in decision-making processes, ethical concerns arise. Certain client segments may be treated unfairly as a result of algorithmic biases (Akter et al., 2021). Businesses must carefully traverse a variety of ethical and legal situations when utilizing AI for fraud detection to maintain ethical standards and assure compliance.

Data quality, system integration, the evolving threat landscape, and regulatory considerations all demand careful attention. By proactively addressing these challenges and fostering a culture of continuous learning and adaptation, organizations can leverage the power of AI to create robust and ethical fraud detection systems, safeguarding their digital operations and fostering trust with their customers.

9. The future of AI and machine learning in redefining financial crime prevention in Nigeria

As we move forward, advancements in AI and ML technologies hold immense promise for creating a more robust and secure financial ecosystem. Here we explore the exciting possibilities that lie ahead in the future of AI and ML in financial crime prevention.

AI systems are capable of constant learning and evolution, in contrast to conventional fraud detection techniques that depend on static rules. AI models may continuously improve their detection capabilities by being exposed to massive volumes of data, including past fraud attempts, new threats, and even successful fraudulent behaviors (Hassan et al., 2023). This guarantees that the system remains innovative, adjusting to the ever-evolving strategies employed by con artists. AI systems that can recognize not only conventional money laundering schemes but also cutting-edge methods that take advantage of newly available financial goods or social engineering advances will be available in the near future (Bello and Olufemi, 2024).

Transformers, known for their success in natural language processing (NLP) and sequence modelling, are emerging as transformative tools in fraud detection. Their ability to model sequential dependencies makes them ideal for identifying fraud patterns in transaction sequences and detecting sophisticated schemes, such as synthetic identity fraud and collusion. Transformers provide deeper insights into complex typologies, helping organizations stay ahead of increasingly sophisticated threats (Vaswani et al., 2017; Wang et al., 2023).

More than just responding to current threats is in store for AI and ML in the future. The potential for proactive fraud prevention exists in many domains of advancement. AI models can forecast, with growing accuracy, the location and timing of fraud attempts by sifting through enormous volumes of previous data, looking for trends linked to fraudulent activity (Martins and Fonkem, 2024). With the help of this predictive capability, businesses can avert fraudulent acts before they do harm by enforcing stricter security procedures or putting in place real-time transaction monitoring (Bansal et al., 2024). An artificial intelligence system that can recognize suspicious transactions based on historical data and emerging patterns and forecast which user profiles or accounts are most susceptible to future fraud attempts is one example.

In addition, a more comprehensive method of fraud detection is anticipated as AI and ML become more prevalent in financial crime prevention. Nowadays, several organizations frequently concentrate on their own internal data while conducting fraud detection operations in silos (Gade, 2024). These silos, however, can be dismantled by AI-powered solutions, which also make it possible to analyze data from a wide range of sources, including social media platforms, banks, telecom companies, and government agencies (Ola-Oluwa, 2024). A more comprehensive understanding of fraudulent activity will be provided by this cooperative approach, enabling law enforcement to more successfully locate and take down intricate criminal networks.

The fusion of AI and other technologies to build a more complete security ecosystem is another fascinating trend. For example, blockchain technology provides an immutable record of transactions, which makes it an ideal match for AI-driven fraud detection systems (Taher et al., 2024). Also, quantum computing introduces the potential to solve complex fraud detection challenges by rapidly analyzing vast datasets and uncovering intricate patterns that classical computing struggle to process using AI (Innan et al., 2024). Through the integration of these technologies, organizations can establish a more secure environment that not only identifies fraudulent actions but also stops them entirely and rapidly. The likelihood of fraudulent transactions being completed can be considerably decreased by this smooth combination of blockchain, quantum computing and AI.

Finally, the future holds promise for the development of more sophisticated AI algorithms. These algorithms will be capable of not only detecting existing fraud typologies but also identifying entirely new ones as they emerge. This will be fueled by the ever-increasing volume and variety of data available for AI models to learn from (Bello and Olufemi, 2024). Imagine an AI system that can not only recognize traditional phishing attempts based on email language patterns but can also detect novel social engineering tactics that exploit emerging social media trends or leverage deepfakes to impersonate legitimate entities. By staying ahead of the curve and continuously adapting to new threats, AI will become a powerful weapon in the fight against financial crime.

There is enormous potential for AI and machine learning to combat financial crime in Nigeria in the future. AI provides an impressive toolkit that includes sophisticated algorithms capable of identifying new fraud schemes, interacting with existing security systems, and proactive protection. Nigeria can protect its people and enterprises in the digital era by embracing these innovations and encouraging cross-sector collaboration to build a stronger and more secure financial environment.

10. Conclusion

Fraud across various sectors in Nigeria is a major economic threat, and traditional methods are failing due to constantly evolving fraud schemes and the vast amount of data. This paper explores how Artificial Intelligence (AI) and Machine Learning (ML) can revolutionize fraud detection. AI and ML excel at recognizing patterns in large datasets, enabling real-time anomaly detection and proactive measures. The benefits include improved accuracy, adaptability, and efficient data handling. Challenges like data quality and ethical considerations exist, but advancements in AI and ML algorithms, integration with technologies like blockchain, and a focus on prevention hold immense promise for the future. By embracing AI and ML responsibly, collaborating across sectors, and prioritizing ethics, Nigeria can create a more secure digital environment that safeguards its citizens and businesses.

To build a safer digital future in Nigeria, several key recommendations are essential. Firstly, financial institutions, governments, and businesses should invest in AI and ML for fraud detection. Secondly, robust data security practices and adherence to privacy laws are crucial for building trust in AI. Thirdly, cross-sector collaboration between banks, telecoms, and government agencies is needed to fight complex financial crimes. Finally, public awareness campaigns can educate citizens about online safety and evolving fraud tactics, empowering them to protect themselves.

CRediT authorship contribution statement

Erastus Olarenwaju Ogunti: Supervision, Conceptualization. **Osekhonmen Victory Abhulimen:** Writing – review & editing, Validation, Project administration, Formal analysis. **Oluwaseun Isaac Odufisan:** Writing – original draft, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI And AI-Assisted Technologies in the Writing Process

During the preparation of this work the author(s) used ChatGPT in order to improve language and readability ONLY. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

Abhulimen, O., Ogunti, E., 2021. Facial age estimation using deep learning: a review. methods 8 (5) http://www.jmest.org/wp-content/uploads/JMESTN42353774.pdf.
Abhulimen, V.O., Ogunti, O.E., 2022. Automatic age estimation of persons with dark skin tone using deep learning approach. Int. J. Comput. Digit. Syst. 12 (1), 1183–1189. https://doi.org/10.12785/ijcds/120194
Adewumi, B., 2024. Nigeria: Artificial Intelligence to push E-Commerce fraud to &107bln. Niger. Trib https://www.zawya.com/en/economy/africa/nigeria-artificial-intelligence-to-push-ecommerce-fraud-to-107bln-xvcqts0q (Accessed 25 October 2024).
Adhikari, P., Hamal, P., Baidoo, F., 2024. Artificial Intelligence in fraud detection: revolutionizing financial security. Int. J. Sci. Res. Arch. 13 (01), 1457–1472. https://doi.org/10.30574/ijsra.2024.13.1.1860
Afriyie, J.K., Tawiah, K., Pels, W.A., Addai-Henne, S., Dwamena, H.A., Owiredu, E.O., Ayeh, S.A., Eshun, J., 2023. A supervised machine learning algorithm for detecting and predicting fraud in credit card transactions. Decis. Anal. J. 6, 100163. https://doi.org/10.1016/j.dajour.2023.100163

Akinniyi, A.J., Erinsakin, O.A., Emma-Ayire, S.O., 2021. Corruption in the Nigerian education sector: causes and remedies. KIU Interdiscip. J. Humanit. Soc. Sci. 2 (2), 1–13.

Akintaro, S., 2023. Banks' losses to frauds hit N5.79 billion in Q2 2023 – FITC – Nairametrics. https://nairametrics.com/2023/08/24/banks-losses-to-frauds/ (Accessed 18 March 2024).

Akokuwebe, M.E., Idemudia, E.S., 2023. Fraud within the Nigerian health system, a double threat for resilience of a health system and the response to the COVID-19 pandemic: a review. Pan Afr. Med. J. 45 (1) https://www.ajol.info/index.php/pamj/article/view/266454/251429.

Akter, S., McCarthy, G., Sajib, S., Michael, K., Dwivedi, Y.K., D'Ambra, J., Shen, K.N., 2021. Algorithmic bias in data-driven innovation in the age of AI. Int. J. Inf. Manag. 60, 102387. https://doi.org/10.1016/j.ijinfomgt.2021.102387

Ali, H.I.H., Alhassan, A., 2021. Fighting contract cheating and ghostwriting in Higher Education: Moving towards a multidimensional approach. Cogent Educ. 8 (1), 1885837. https://doi.org/10.1080/2331186X.2021.1885837

Alordiah, C., Omumu, F., Kaizar, V., 2023. Investigating why students in Nigeria perceive education as a scam. J. Appl. Learn. Teach. 6 (1), 234–243 https://journals.sfu.ca/jalt/index.php/jalt/article/download/685/589.

Amiri, Z., Heidari, A., Navimipour, N.J., Unal, M., Mousavi, A., 2024. Adventures in data analysis: a systematic review of Deep Learning techniques for pattern recognition in cyber-physical-social systems. Multimed. Tools Appl. 83 (8), 22909–22973. https://doi.org/10.1007/s11042-023-16382-x

Arslan, B., Yorulmaz, E., Akca, B., Sagiroglu, S., 2016. Security Perspective of Biometric Recognition and Machine Learning Techniques, pp. 492–497. https://doi.org/10.1109/ICMLA.2016.0087.

AxiomQ, 2022. 6 Types Of eCommerce Fraud and How To Prevent Them. Retrieved from: https://axiomq.com/blog/6-types-of-ecommerce-fraud-and-how-to-prevent-them/ (Accessed 18 March 2024).

Bansal, U., Bharatwal, S., Bagiyam, D.S., Kismawadi, E.R., 2024. Fraud detection in the era of AI: Harnessing technology for a safer digital economy. In: AI-Driven Decentralized Finance and the Future of Finance. IGI Global, pp. 139–160. https://doi.org/10.4018/979-8-3693-6321-8.ch006

Banwo&Ighodalo, 2023. Nigeria Data Protection Act: What Individuals, Businesses And Organizations Should Know. https://banwo-ighodalo.com/assets/grey-matter/908710c62e3fce362d559eaf7ee7932f.pdf (Accessed 10 October 2024).

Bello, O.A., Folorunso, A., Ogundipe, A., Kazeem, O., Budale, A., Zainab, F., Ejiofor, O.E., 2022. Enhancing cyber financial fraud detection using deep learning techniques: a study on neural networks and anomaly detection. Int. J. Netw. Commun. Res. 7 (1), 90–113 https://tudr.org/id/eprint/3107.

Bello, H.O., Ige, A.B., Ameyaw, M.N., 2024. Adaptive machine learning models: concepts for real-time financial fraud prevention in dynamic environments. World J. Adv. Eng. Technol. Sci. 12 (02), 021–034. https://doi.org/10.30574/wjaets.2024.12.2.0266

Bello, O.A., Olufemi, K., 2024. Artificial intelligence in fraud prevention: exploring techniques and applications challenges and opportunities. Comput. Sci. IT Res. J. 5 (6), 1505–1520. https://doi.org/10.51594/csitrj.v5i6.1252

Boonkrong, S., 2024. Design of an academic document forgery detection system. Int. J. Inf. Technol. 1–13. https://doi.org/10.1007/s41870-024-02006-6

Castelblanco, A., Solano, J., Lopez, C., Rivera, E., Tengana, L., Ochoa, M., 2020. Machine Learning Techniques for Identity Document Verification in Uncontrolled Environments: A Case Study. pp. 271–281. https://doi.org/10.1007/978-3-030-49076-8_26.

Challoumis, C., 2024. the Future of Business-integrating AI Into the Financial Cycle. XIV Int. Sci. Conf. 44–78.

Chen, J., Lai, K.L., 2021. Deep convolution neural network model for credit-card fraud detection and alert. J. Artif. Intell. Capsul. Netw. 3 (1), 101–112. https://doi.org/10.36548/jaicn.2021.2.003

Drammeh, F., 2023. Trust and Fraud in Nigeria: A Comprehensive Analysis of Socioeconomic Factors and Regulatory Measures. https://dx.doi.org/10.2139/ssrn.4475135.

Emiroğlu, E.E., 2023. Texture Analysis and Classification by Deep Architectures for Paper Fraud Detection (Doctoral dissertation, Middle East Technical University (Turkey)). https://open.metu.edu.tr/bitstream/handle/11511/105954/thesis.pdf.

Eneji, C.V.O., Petters, J.S., Esuabana, S.B., Onnoghen, N.U., Udumo, B.O., Ambe, B.A., Essien, E.E., Unimna, F.A., Alawa, D.A., Ikutal, A., 2022. University academic dishonesty and graduate quality for national development and global competitiveness: Nigerian universities in perspective. Int. J. Learn., Teach. Educ. Res. 21 (5), 401–427. https://doi.org/10.26803/ijlter.21.5.20

Fraud.net, 2024. How to use machine learning for fraud detection and prevention. https://fraud.net/n/how-to-use-machine-learning-for-fraud-detection-and-prevention/ (Accessed 19 March 2024).

Gade, K.R., 2024. Beyond data quality: building a culture of data trust. J. Comput. Inf. Technol. 4 (1) https://universe-publisher.com/index.php/jcit/article/download/5/5.

Gayam, S.R., 2020. AI-driven fraud detection in E-commerce: advanced techniques for anomaly detection, transaction monitoring, and risk mitigation. Distrib. Learn. Broad Appl. Sci. Res. 6, 124–151 https://dlabi.org/index.php/journal/article/download/108/105.

Goyal, M.A., Singh, M.S., Sharma, E.S., 2020. Fraud detection on social media using data analytics. Int. J. Eng. Res. Technol. 9. https://doi.org/10.17577/IJERTV9IS010204

Guan, C., Suryanto, H., Mahidadia, A., Bain, M., Compton, P., 2023. Responsible credit risk assessment with machine learning and knowledge acquisition. Hum. -Centr Intell. Syst. 3 (3), 232–243. https://doi.org/10.1007/s44230-023-00035-1

Gupta, P., 2023. Leveraging machine learning and artificial intelligence for fraud prevention. SSRG Int. J. Comput. Sci. Eng. 10 (5), 47–52. https://doi.org/10.14445/23488387/IJCSE-V10I5P107

Hassan, M., Aziz, L.A.R., Andriansyah, Y., 2023. The role artificial intelligence in modern banking: an exploration of AI-driven approaches for enhanced fraud prevention, risk management, and regulatory compliance. Rev. Contemp. Bus. Anal. 6 (1), 110–132 https://researchberg.com/index.php/rcba/article/download/153/148.

Hilal, W., Gadsden, S.A., Yawney, J., 2022. Financial fraud: a review of anomaly detection techniques and recent advances. Expert Syst. Appl. 193, 116429. https://doi.org/10.1016/j.eswa.2021.116429. https://www.grrbe.in/Vol-10-issue-05/GRRBE_L_100100910201121320.pdf.

Innan, N., Sawaika, A., Dhor, A., Dutta, S., Thota, S., Gokal, H., ... Bennai, M., 2024. Financial fraud detection using quantum graph neural networks. Quant. Mach. Intell. 6 (1), 7. https://doi.org/10.1007/s42484-024-00143-6.

Jóhannsson, J., 2017. Detecting fraudulent users using behaviour analysis. [Online]. Available: https://api.semanticscholar.org/CorpusID:79756.

Juniper Research, 2022. Online Payment Fraud Losses to Exceed $343 Billion Globally. https://www.juniperresearch.com/press/online-payment-fraud-losses-to-exceed-343bn/ (Accessed 17 March 2024). https://www.juniperresearch.com/press/online-payment-fraud-losses-to-exceed-343bn/ (Accessed 17 March 2024).

Kamuangu, P., 2024. A Review on Financial Fraud Detection using AI and Machine Learning. J. Econ. Financ. Account. Stud. 6 (1), 67–77. https://doi.org/10.32996/jefas.2024.6.1.7

Kasaraneni, B.P., 2024. Machine learning models for fraud detection in health insurance claims: techniques, applications, and real-world case studies. J. Mach. Learn. Pharm. Res. 4 (1), 110–147 https://pharmapub.org/index.php/jmlpr/article/download/42/40.

Khurana, R., 2020. Fraud detection in ecommerce payment systems: The role of predictive ai in real-time transaction security and risk management. Int. J. Appl. Mach. Learn. Comput. Intell. 10 (6), 1–32 https://neuralslate.com/index.php/Machine-Learning-Computational-I/article/view/155.

Komandla, V., 2024. Enhancing Security and Fraud Prevention in Fintech: Comprehensive Strategies for Secure Online Account Opening. Glob. Res. Rev. Bus. Econ. 10 (5), 102–112.

Li, J., Li, N., Xia, T., Guo, J., 2023. Textual analysis and detection of financial fraud: Evidence from Chinese manufacturing firms. Econ. Model. 126, 106428. https://doi.org/10.1016/j.econmod.2023.106428

Lin, A.K., 2024. The AI revolution in financial services: emerging methods for fraud detection and prevention. J. Galaksi 1 (1), 43–51. https://doi.org/10.70103/galaksi.v1i1.5

Liu, Q., Hagenmeyer, V., Keller, H.B., 2021. A review of rule learning-based intrusion detection systems and their prospects in smart grids. IEEE Access 9, 57542–57564. https://doi.org/10.1109/ACCESS.2021.3071263

Lokanan, M., Tran, V., Vuong, N.H., 2019. Detecting anomalies in financial statements using machine learning algorithm: the case of Vietnamese listed firms. Asian J. Account. Res. 4 (2), 181–201. https://doi.org/10.1108/AJAR-09-2018-0032

Lu, Q., Fu, C., Nan, K., Fang, Y., Xu, J., Liu, J., Bellotti, A.G., Lee, B.G., 2023. Chinese corporate fraud risk assessment with machine learning. Intell. Syst. Appl. 20, 200294. https://doi.org/10.1016/j.iswa.2023.200294

Lu, J., Lin, K., Chen, R., Lin, M., Chen, X., Lu, P., 2023. Health insurance fraud detection by using an attributed heterogeneous information network with a hierarchical attention mechanism. BMC Med. Inform. Decis. Mak. 23 (1), 62. https://doi.org/10.1186/s12911-023-02152-0

Malik, P., Anand, A., Baliyan, A.K., Dongre, A., Panwar, P., 2024. Credit risk assessment and fraud detection in financial transactions using machine learning. J. Electr. Syst. 20, 2061–2069. https://doi.org/10.52783/jes.1807

Manyo, T.S., Walter, M.H., Bassey, O.E., Wonah, O.G., Omang, B.A., Ekpo, N.S., Chike, E.C., 2023. Effect of fraud on commercial banks' performance in Nigeria. Front. Manag. Sci. 2 (2), 69–78. https://doi.org/10.56397/FMS.2023.04.08

Martins, O., Fonkem, B., 2024. Leveraging big data analytics to combat emerging financial fraud schemes in the USA. A Lit. Rev. Pract. Implic. https://doi.org/10.30574/wjarr.2024.24.1.2999

Mothukuri, V., Parizi, R.M., Pouriyeh, S., Huang, Y., Dehghantanha, A., Srivastava, G., 2021. A survey on security and privacy of federated learning. Future Gener. Comput. Syst. 115, 619–640. https://doi.org/10.1016/j.future.2020.10.007

Mutemi, A., Bacao, F., 2024. E-commerce fraud detection based on machine learning techniques: systematic literature review. Big Data Min. Anal. 7 (2), 419–444 https://ieeexplore.ieee.org/iel7/8254253/10506765/10506811.pdf.

Nagpal, N., Srivastava, A., Verma, P., 2024. AI-powered proctoring: safeguarding online assessment in the education 5.0. In: Explainable AI for Education: Recent Trends and Challenges. Springer Nature Switzerland, Cham, pp. 271–285. https://doi.org/10.1007/978-3-031-72410-7_15

Nnamoko, N., Barrowclough, J., Liptrott, M., Korkontzelos, I., 2022. A behaviour biometrics dataset for user identification and authentication. Data Brief. 45, 108728. https://doi.org/10.1016/j.dib.2022.108728

Noah, A.O.K., Adesope, J.A., 2022. Phenomenon of Corrupt Practices: Admission racketeering by Staff in Public Tertiary Institutions in Lagos. Nigeria, pp. 194–208.

Okechukwu, O.P., Okechukwu, G.N., Mbonu, C.E., Paul, R.U., 2023. A deep learning model for detecting anomalies in the banking sector using a feed-forward neural network. Int. J. Sci. Eng. Res. 14, 322–327 ISSN 2229-5518.

Okolie, U.C., Ojomo, A.H., 2020. E-commerce in Nigeria: benefits and challenges. Humanit. Soc. Sci. Latv. 28 (2). https://doi.org/10.22364/hssl.28.2.05

Okonkwo, N.O., Idika, N.K., Kalu, S.A., 2024. DIgital economy and its implications for sustainable economic growth in Nigeria. Adv. J. Arts, Humanit. Soc. Sci. 7 (3), 40–53 https://aspjournals.org/ajahss/index.php/ajahss/article/view/100.

Okoye, E.I., Gbegi, D.O., 2013. An evaluation of the effect of fraud and related financial crimes on the Nigerian economy. 856 (33), 1–23. https://doi.org/10.12816/0001221

Oladimeji, O.O., 2019. Text analysis and machine learning approach to phished email detection. Int. J. Comput. Appl. 182 (36). https://doi.org/10.5120/ijca2019918354

Ola-Oluwa, J.A., 2024. Impact of Artificial Intelligence (AI) in enhancing knowledge sharing and boosting organizational efficiency in Nigerian enterprises. Afr. J. Manag. Bus. Res. 17 (1), 76–95. https://doi.org/10.62154/ajmbr

Oloni, V., 2024. From Brick to Click: E-commerce and the Future of Retail. https://www.verivafrica.com/insights/from-brick-to-click-e-commerce-and-the-future-of-retail (Accessed 24 October 2024).

Oluwole, V., 2024. 4 African countries with highest scam losses in 2024. Business Insider Africa. https://africa.businessinsider.com/local/markets/african-countries-with-highest-scam-losses/wrvpj5k (Accessed 15 December 2024).

Paul, H., Nikolaev, A., 2021. Fake review detection on online E-commerce platforms: a systematic literature review. Data Min. Knowl. Discov. 35 (5), 1830–1881. https://doi.org/10.1007/s10618-021-00772-6

Phua, C., Lee, V., Smith, K., Gayler, R., 2010. A comprehensive survey of data mining-based fraud detection research. arXiv Prepr. arXiv 1009.6119.

Pratik, S., Ashwin, Suresh, G., Sushant, I., 2020. Anomaly detection using machine learning. Int. J. Innov. Sci. Res. Technol. (IJISRT 5 (3), 1458–1461 https://bit.ly/2ycrxti.

Rai, A.K., Dwivedi, R.K., 2020. Fraud detection in credit card data using unsupervised machine learning based scheme. In: 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC). IEEE, pp. 421–426. https://doi.org/10.1109/icesc48915.2020.9155615

Reurink, A., 2018. Financial fraud: a literature review. J. Econ. Surv. 32 (5), 1292–1325. https://doi.org/10.1111/joes.12294

Sadgali, I., Sael, N., Benabbou, F., 2021. Human behavior scoring in credit card fraud detection. IAES Int. J. Artif. Intell. 10 (3), 698. https://doi.org/10.11591/ijai.v10.i3.pp698-706

Sarker, I.H., 2021. Machine learning: algorithms, real-world applications and research directions. SN Comput. Sci. 2, 160. https://doi.org/10.1007/s42979-021-00592-x

Sharma, R., Mehta, K., Sharma, P., 2024. Role of artificial intelligence and machine learning in fraud detection and prevention. In: Risks and Challenges of AI-Driven Finance: Bias, Ethics, and Security. IGI Global, pp. 90–120. https://doi.org/10.4018/979-8-3693-2185-0.ch005

Taher, S.S., Ameen, S.Y., Ahmed, J.A., 2024. Advanced fraud detection in blockchain transactions: an ensemble learning and explainable AI approach. Eng. Technol. Appl. Sci. Res. 14 (1), 12822–12830. https://doi.org/10.48084/etasr.6641

Takyar, A., 2024. Financial fraud detection using machine learning models. https://www.leewayhertz.com/build-financial-fraud-detection-system-using-ML-models/ (Accessed 19 March 2024).

Theodorakopoulos, L., Theodoropoulou, A., Stamatiou, Y., 2024. A state-of-the-art review in big data management engineering: real-life case studies, challenges, and future research directions. Eng 5 (3), 1266–1297. https://doi.org/10.3390/eng5030068

Thornton, D., Brinkhuis, M., Amrit, C., Aly, R., 2015. Categorizing and describing the types of fraud in healthcare. Procedia Comput. Sci. 64, 713–720. https://doi.org/10.1016/j.procs.2015.08.594

Tripathi, A., Thakar, S.V., 2024. Ethical use of AI for academic integrity: preventing plagiarism and cheating. Ethic Framew. Spec. Educ.: A Guide Res. 91.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst https://user.phil.hhu.de/~cwurm/wp-content/uploads/2020/01/7181-attention-is-all-you-need.pdf.

Wang, H., Zheng, J., Carvajal-Roca, I.E., Chen, L., Bai, M., 2023, August, August. Financial fraud detection based on deep learning: towards large-scale pre-training transformer models. In: China Conference on Knowledge Graph and Semantic Computing. Springer Nature Singapore, Singapore, pp. 163–177. https://doi.org/10.1007/978-981-99-7224-1_13

Yange, T.S., 2019. A fraud detection system for health insurance in Nigeria. J. Health Inform. Afr. 6 (2), 64–73. https://doi.org/10.12856/JHIA-2019-v6-i2-256

Yousefi, N., Alaghband, M., Garibay, I., 2019. A comprehensive survey on machine learning techniques and user authentication approaches for credit card fraud detection. Retrieved from: arXiv preprint arXiv:1912.02629.